

☒ Easy Level (10 Questions)

Q1. How do you define a function with an optional argument msg defaulting to "Hello"?

- A. `def f(msg="Hello"):`
- B. `def f(msg):`
- C. `def f(msg=None):`
- D. `def f(msg!="Hello"):`

Answer: A

Explanation: Default arguments are assigned as `arg=default`.

Q2. What does `type(10)` return?

- A. `'int'`
- B. `<class 'int'>`
- C. `'10'`
- D. `int`

Answer: B

Explanation: `type()` returns the class of the object.

Q3. What is the purpose of `dir(obj)`?

- A. Returns the type of `obj`
- B. Returns list of attributes and methods of `obj`
- C. Converts `obj` to string

D. Checks if obj has an attribute

Answer: B

Explanation: dir() lists available attributes/methods.

Q4. What will this print?

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```
def greet(name, msg="Hello"): print(f'{msg}, {name}!') greet("Komal")
```

A. Hello, Komal!

B. , Komal!

C. Hello, !

D. Error

Answer: A

Explanation: Default value of msg is used.

Q5. Which built-in function returns a string representation?

- A. `str()`
- B. `type()`
- C. `dir()`
- D. `id()`

Answer: A

Explanation: `str()` converts to string.

Q6. What does `id(obj)` return?

- A. Object's memory address identifier
- B. Object's type
- C. Object's string representation
- D. Number of attributes

Answer: A

Explanation: `id()` returns unique identifier.

Q7. If `a = [1, 2]` and `b = a`, what happens if `b.append(3)`?

- A. Only `b` changes
- B. Both `a` and `b` change
- C. Error
- D. `a` resets

Answer: B

Explanation: Both reference same list.

Q8. What module is used for network sockets in Python?

- A. network
- B. socket
- C. net
- D. tcp

Answer: B

Explanation: socket is the built-in module.

Q9. Which socket type is used for TCP?

- A. SOCK_DGRAM
- B. SOCK_STREAM
- C. AF_INET
- D. AF_UNIX

Answer: B

Explanation: TCP uses stream sockets.

Q10. Which method binds a socket to an address?

- A. listen()
- B. accept()
- C. connect()
- D. bind()

Answer: D

Explanation: bind() assigns socket to IP and port.

☒ Medium Level (20 Questions)

Q11. Which is the correct way to call function with named arguments?

A. `f(msg="Hi", name="Komal")`

B. `f("Komal", msg="Hi")`

C. `f(name="Komal", "Hi")`

D. Both A and B

Answer: D

Explanation: Positional args before named, both shown are valid.

Q12. What does this print?

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```
print(str([1, 2, 3]))
```

- A. [1, 2, 3]
- B. '1, 2, 3'
- C. [1, 2, 3] as list
- D. Error

Answer: B

Explanation: str() returns string form including brackets.

Q13. What will dir(5) return?

- A. List of int methods and attributes
- B. List of int values
- C. String 'int'
- D. Error

Answer: A

Explanation: dir() lists methods for int object.

Q14. What is printed here?

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```
def f(x, y=10): print(x, y) f(5)
```

- A. 5 10
- B. 10 5
- C. 5 5
- D. Error

Answer: A

Explanation: y defaults to 10.

Q15. How to check if object obj has attribute attr?

- A. hasattr(obj, 'attr')
- B. obj.has('attr')
- C. obj.attr
- D. attr in obj

Answer: A

Explanation: hasattr() checks attribute existence.

Q16. What happens when `id(a) == id(b)` is True?

- A. a and b reference same object
- B. a and b have same value but different objects
- C. a is copy of b
- D. Error

Answer: A

Explanation: Same id means same object.

Q17. Which method listens for connections on a socket?

- A. connect()
- B. listen()
- C. send()
- D. accept()

Answer: B

Explanation: listen() enables socket to accept clients.

Q18. What does socket.AF_INET represent?

- A. IPv4 address family
- B. IPv6 address family
- C. UNIX sockets
- D. Stream sockets

Answer: A

Explanation: AF_INET means IPv4.

Q19. Which function receives data from a socket?

- A. recv()
- B. send()

C. connect()

D. bind()

Answer: A

Explanation: recv() reads incoming data.

Q20. What is the difference between send() and sendall()?

A. send() may not send all bytes, sendall() sends all

B. No difference

C. send() sends strings, sendall() sends bytes

D. send() closes connection

Answer: A

Explanation: sendall() ensures all data is sent.

🚩 Hard Level (10 Questions)

Q21. What will this print?

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```
def f(a, b=2, *, c=3): print(a, b, c) f(1, c=4)
```

A. 1 2 4

B. 1 4 3

C. 1 3 4

D. Error

Answer: A

Explanation: c is keyword-only argument; b defaults to 2.

Q22. What does this output?

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```
a = [1, 2, 3] b = a[:] b.append(4) print(a, b)
```

- A. [1, 2, 3] [1, 2, 3, 4]
- B. [1, 2, 3, 4] [1, 2, 3, 4]
- C. [1, 2, 3] [1, 2, 3]
- D. Error

Answer: A

Explanation: b is a shallow copy; a remains unchanged.

Q23. Which built-in function helps see an object's documentation?

- A. help()
- B. doc()
- C. info()
- D. describe()

Answer: A

Explanation: help() shows documentation.

Q24. Which of the following socket calls is blocking?

- A. connect()
- B. send()
- C. accept()
- D. All of the above

Answer: D

Explanation: These calls block until operation completes.

Q25. What is the result of this?

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```
def f(x, y=1, *args, **kwargs): print(x, y, args, kwargs) f(5, 6, 7, 8, a=9)
```

- A. 5 6 (7, 8) {'a': 9}
- B. 5 1 (6, 7, 8) {'a': 9}
- C. 5 6 7 {'a': 9}
- D. Error

Answer: A

Explanation: *args collects extra positional, **kwargs collects named args.

Q26. What does isinstance(obj, int) check?

- A. If obj is exactly int type
- B. If obj is an instance or subclass of int
- C. If obj can be cast to int
- D. None

Answer: B

Explanation: Checks inheritance too.

Q27. What does this do?

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```
s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
```

- A. Creates a UDP socket
- B. Creates a TCP socket
- C. Creates a UNIX socket
- D. Error

Answer: A

Explanation: SOCK_DGRAM is UDP.

Q28. What is the purpose of `socket.close()`?

- A. Closes the socket connection

- B. Opens a new socket
- C. Sends data
- D. Listens for clients

Answer: A

Explanation: `close()` releases socket resource.

Q29. How can you ensure a Python socket is non-blocking?

- A. `sock.setblocking(False)`
- B. `sock.nonblocking(True)`
- C. `sock.block(False)`
- D. `sock.setblock(False)`

Answer: A

Explanation: `setblocking(False)` sets socket non-blocking.

Q30. What does the * symbol in function parameters mean?

- A. Positional-only arguments
- B. Keyword-only arguments after *
- C. Variable number of positional arguments
- D. Multiply arguments

Answer: B

Explanation: * marks start of keyword-only arguments.